

1. INTRODUCTION

Our team developed a multimodal skin lesion classifier that integrates dermoscopic images with tabular patient data to categorize skin lesions into seven classes spanning malignant, premalignant, and benign conditions. The dataset's heavy class imbalance drove us to focus on correctly classifying minority classes, with macro F1 as our primary evaluation metric. We explored several modeling approaches, settling on an intermediate fusion architecture consisting of ResNet for image feature extraction and an MLP for tabular data features, followed by a linear layer for prediction on the concatenated image and tabular data. Image preprocessing consisted of border cropping, hair removal, color constancy, image augmentation, and ResNet image normalization. Based on the public leaderboard results, our model had the third best performance with an F1 score of 0.585. Post-hoc analysis of our model's confusion matrix, t-SNE projection of embeddings, and Grad-CAM images were performed and discussed below.

2. METHODS

2.1 - Model Architecture

Three model architectures were tested (Figure 1). We began with a late fusion approach combining probabilities from a CNN image classifier (ReLU, max pooling, MLP head) and a

gradient-boosting tabular data classifier with hyperparameter tuning. Probabilities were averaged via experimentally-determined weighting to obtain predictions. We selected a CNN for its spatial feature extraction and efficiency on image datasets and gradient boosting for its strong performance on tabular data. Test F1 was 0.202, and we hypothesized that the model may benefit from learning interactions between modalities at the feature level rather than the decision level.

Thus, we tried an intermediate fusion model: a three-layer CNN for the image data, an MLP for the tabular data, and a linear layer with softmax that classifies the concatenated image and tabular features. This model did not have substantially better performance than the late fusion model. To address this, we implemented Resnet-18 into this intermediate fusion model (He et al., 2015). The pretrained backbone allowed the model to leverage the weights learned from the large ImageNet dataset while also fine-tuning based on the skin lesion images, rather than learning from scratch. Also, ResNet's greater depth allowed the model to learn more complex features than our shallow CNN, while reducing the risk of vanishing gradients via skip connections.

2.2 - Preprocessing Pipeline

Border cropping was introduced to remove non-skin image backgrounds, including gray borders (see first slide of Figure 2) or microscope/camera lens artifacts. This was done by center-cropping the 224x224 image to 204x204, then padding the borders to restore the original image size. This was intentional as ResNet was trained on 224x224 images. When padding the borders, the median color value of

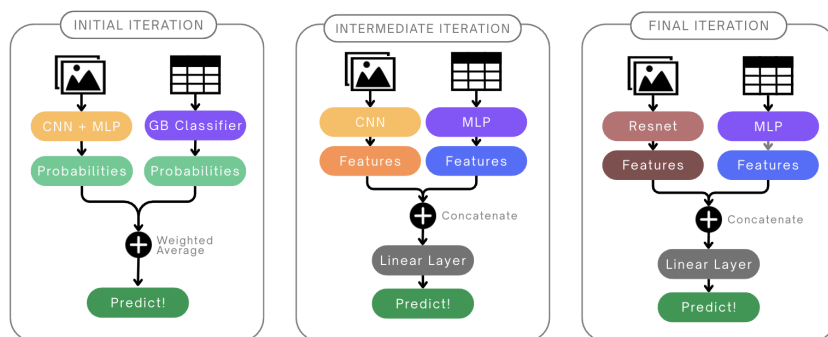


Figure 1: Implemented model architectures

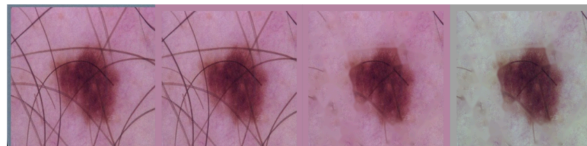


Figure 2: Example of preprocessing pipeline

the outer 20 pixels of the image was used to fill in the borders; we found that this was sufficient to capture the true skin color rather than the non-skin artifacts.

Hair removal was motivated by our brief literature review of deep learning-based skin lesion classification, which suggested that hair removal is beneficial as it creates a cleaner image for analysis (Francode77, 2023; Mustafa et al., 2025). Our hair removal approach utilized grayscaling, edge detection, and cv2inpaint to identify, mask, and fill in the hairs based on the surrounding image color, as seen in Figure 2.

Color constancy was also implemented to remove changes in color from varying lighting conditions by normalizing color and illumination (Arora, 2020; Barata et al., 2015). Our approach was based on an open-source implementation (Rana, 2018). This approach rescales the image to enforce standard grayscale coloring based on the underlying assumption that under neutral lighting conditions, the average color would be gray.

Image augmentation, consisting of random horizontal and vertical flips, was applied to improve model robustness. The final image preprocessing step was image normalization using the mean and standard deviation values from the ImageNet dataset on which ResNet was trained to improve model convergence and stability.

To preprocess the tabular data, we used an sklearn ColumnTransformer to apply one-hot encoding on the categorical data (sex and anatomical site) and standardize the age data separately. These steps made little difference to our results but were intended to ensure the model was ignoring ordinal relationships in the categorical data and could easily interpret age.

2.3 - Model Training Design Choices

To address some overfitting noticed in earlier iterations of the model, we used AdamW with a smaller learning rate for the pretrained CNN ($1e-5$), and a larger learning rate for the tabular MLP and final classifier ($1e-4$), as well as weight decay to encourage weights to stay small.

Weighted cross-entropy was implemented to address the heavy class imbalance by penalizing errors on minority classes. Despite improved F1 performance after these changes, the model was still heavily biased towards predicting the majority class. As a result, we also applied focal loss (Yeung et al., 2022), which downweights the loss of easy samples, forcing the model to focus on harder-to-classify samples.

Preliminary results, while improved, still showed somewhat unstable training loss and performance gaps between the training, validation, and test datasets, suggesting the model continued to overfit. To address this, we added a learning rate scheduler that halved the learning rate whenever F1 plateaued across three epochs.

3. EXPERIMENT SETUP

We used an 80/20 training-validation split on the given training data, stratified to maintain the distribution of the class labels. Next, the preprocessing steps in part (2) were applied. Then, the model was trained for 20 epochs with a batch size of 64 and dropout rate of 0.3. During training, we evaluated on the validation set after each epoch and tracked validation loss, accuracy, and F1. The best model was selected and saved based on the epoch with the highest validation F1.

For test predictions, we again applied the image preprocessing pipeline before loading in the saved model weights to perform predictions with test-time augmentation—separating each image into four copies with different augmentations, running the model on each, and averaging probabilities to get the final prediction.

4. RESULTS

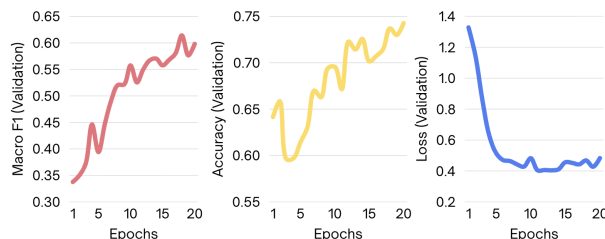


Figure 3: Validation metrics of best-performing model across 20 epochs

Model Iterations	Test F1	Val F1	Val Acc
Late Fusion (CNN + GBC)	0.202	0.227	0.623
Intermediate Fusion (ResNet + MLP)	0.438	0.426	0.876
Add Img. Augmentation & Normalization	0.537	0.607	0.783
Add Preprocessing of Tabular Data	0.485	0.628	0.750
Add Learning Rate Scheduler	0.524	0.606	0.749
Add Focal Loss, Test-Time Augmentation	0.585	0.614	0.737

Table 1: Model performance across iterations

The largest improvements in our test F1 performance came from implementing intermediate fusion with ResNet, increasing the number of epochs, and adding focal loss. While earlier models plateaued around epochs 5-10, adding class weights, weight decay, and a learning rate scheduler allowed the model to continue improving its validation F1 score through 20 epochs by punishing overfitting and forcing the model to pay closer attention to the minority classes.

Validation accuracy also decreased across model iterations, suggesting the model was learning more balanced class patterns rather than overpredicting the dominant class. During training, validation accuracy has sharp spikes, likely reflecting focal loss shifting attention toward harder samples. By intentionally reducing overfitting and accounting for class imbalance, we observed that the delta between validation and test F1 slowly decreased over model iterations.

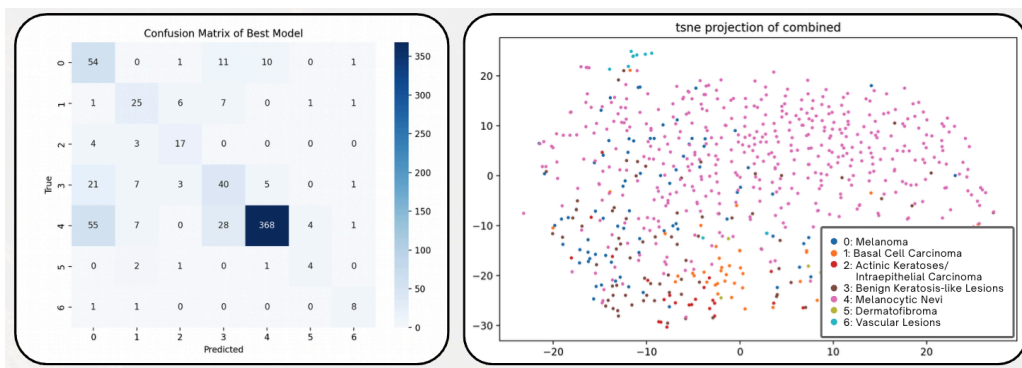


Figure 4: (a) CM of best test F1 model, (b) t-SNE projection of image and feature embeddings

The confusion matrix and t-SNE embeddings show that the model’s feature space is partially separable with significant overlap among some classes (Figure 4). Particularly, class 0 and class 3 seem to overlap significantly with each other in the embedding space, especially in the lower left quadrant of the embedding space (Figure 4b). Furthermore, class 4 is often misclassified as class 0 and 3 (Figure 4a). Examination of the class labels reveals that classes 3 and 4 correspond to benign diagnoses, while class 0 is cancerous. When looking deeper into the classes, we found many guides discussing how to distinguish skin markings that are of class 0 versus class 4, possibly suggesting that our model encounters similar skin diagnosis issues as the general public.

Another stimulating observation was the high rate of misdiagnosis of class 1 as class 2 or class 3 (Figure 4a). Class 1 is malignant, while classes 2 and 3 correspond to a precancer lesion and a benign but cancerous-appearing lesion. These classes also appeared to occupy the lower third of the embedding space together, and the points were relatively sparse and intermixed. It would be interesting to examine whether existing research indicates that these types of misdiagnoses are clinically common. It would also be cool to see whether the class 1 lesions had originally been precancerous class 2.

Finally, we applied Grad-CAM (gradient-weighted class activation mapping) to visualize which regions of the image the model focused on to make the final prediction (Gildenblat et al., 2021). Correctly classified images tended to have weights concentrated around the lesion area, while misclassified cases often had more diffuse or mislocalized focus. However, as such visualization methods can sometimes be misleading—especially due to the depth of the ResNet model—we interpreted these results cautiously as an additional source of information alongside other interpretation methods.

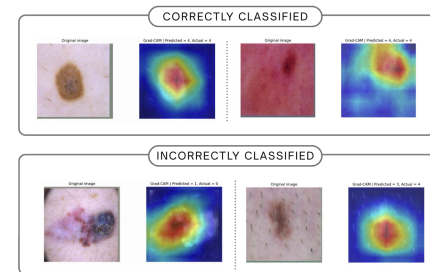


Figure 5: Grad-CAM examples

Overall, our multimodal ResNet + MLP approach improved performance significantly over earlier approaches, achieving fairly strong macro F1. However, performance is still limited by susceptibility to class imbalance and difficulty distinguishing between certain classes, suggesting that stronger approaches to handling performance on minority classes would be the next step.

References

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Contributions

Overall, the work was evenly split among the team. Everyone contributed to each component of the project, though Kelley focused on image preprocessing, model architecture, and model explainability, Russell focused on model implementation and test-time augmentation, and Brandon focused on image preprocessing and results interpretation.

AI Statement

AI was used for idea generation, specifically for information on the differences between Adam versus AdamW optimizers, as well as information on focal versus weighted cross-entropy loss.

Github Repository Link

<https://github.com/kelleytu/cs1851-final-project.git>